



A Stacking Ensemble Approach for Real-Time Anomaly Detection in IoMT Networks Using Healthcare Data

**Dr. Sreeram Indraneel¹, Bandla Krishna Priya², Bonigala Nandini³, Battula Yuva Durga⁴,
Asodi Keerthi Reddy⁵, Shaik Sabeeha⁶**

¹Professor & Head of the Department, St. Ann's College of Engineering & Technology, Chirala,
Andhra Pradesh- 523157

²⁻⁶UG Student, St. Ann's College of Engineering & Technology, Chirala, Andhra Pradesh-
523157

ABSTRACT: The Internet of Medical Things (IoMT) integrates interconnected medical devices to enable real-time monitoring and personalized healthcare, but this connectivity also introduces serious cybersecurity risks such as data falsification, manipulation, and DOS attacks that threaten patient safety and service reliability. The objective of this study is to develop a real-time anomaly detection system capable of identifying malicious activities during IoMT data transmission. Manual monitoring approaches rely heavily on human observation and fixed rule based checking, which are slow, inconsistent, and unable to detect complex cyberattacks, making them unsuitable for high-speed IoMT environments. Previous system implementations using Random Forest, XG Boost, KNN, SVM and Artificial Neural Networks (ANN) provide enhanced detection capabilities but still struggle with processing efficiency, computational load, and adaptability to diverse medical attack scenarios. To address these limitations, this work proposes a machine learning based anomaly detection model using the Light Gradient Boosting Machine (Light GBM) algorithm. A new medical cybersecurity dataset inspired by UNSW-NB15 was developed, containing anomalous traffic and healthcare-specific attack types such as falsification and DoS patterns, providing a realistic and challenging evaluation environment. Light GBM was selected for its high efficiency, reduced computational overhead, and strong performance.

Key Words: IoMT, Anomaly Detection, Machine Learning, Light GBM, Cyber security, Healthcare Dataset.

1. INTRODUCTION

The Internet of Medical Things (IoMT) has revolutionized modern healthcare by seamlessly connecting medical devices, wearable sensors, and healthcare IT systems [1]. This interconnected ecosystem enables real-time patient monitoring, personalized treatment plans, and efficient care delivery, significantly enhancing healthcare quality and patient outcomes. Devices such as smart infusion pumps, wearable health monitors and connected diagnostic tools empower healthcare providers to make timely interventions and deliver tailored care strategies. Beyond clinical benefits, IoMT reduces costs, improves operational efficiency, and increases accessibility to medical services, holding immense potential to redefine healthcare systems globally [1], [2].

A healthcare-specific dataset for anomaly detection in IoMT networks was developed to address the limitations of existing datasets. It incorporates realistic medical scenarios and includes critical cyberattack types such as data falsification and denial-of-service (DoS), inspired by the **UNSW-NB15** dataset. This dataset provides a

reliable benchmark for evaluating the performance of machine learning-based Intrusion Detection Systems (IDS) in healthcare environments. It includes diverse normal and anomalous data, balancing complexity and practicality for IoMT security research. Using this dataset, we evaluated seven ML algorithms, XGBoost, Random Forest, ANN, Support Vector Machines (SVM) [28], K-Nearest Neighbors (KNN), Logistic Regression (LR), and Isolation Forest (IF), and selected XGBoost, Random Forest, and ANN as base learners for their strong accuracy and efficiency. A novel stacking ensemble model was developed for real-time anomaly detection in IoMT environments, using XGBoost as the meta-learner to enhance accuracy and efficiency. The model addresses key cybersecurity challenges such as data integrity and availability by adapting dynamically to incoming data and emerging attack patterns. To support this, a healthcare-specific dataset was created by combining IoT network attacks (e.g., fuzzers, worms, exploits) with healthcare-relevant threats like data falsification and DoS. The proposed approach demonstrates improved

performance, providing a reliable solution for anomaly detection in healthcare systems.

2. LITERATURE SURVEY

The increasing use of machine learning for anomaly detection in IoMT and IoT networks has led to diverse approaches, datasets, and methodologies. This section reviews key contributions, focusing on anomaly detection techniques, ensemble frameworks, and their applications in healthcare and IoT security. A comparison of prior works is provided in Appendix.

In [8], five machine learning algorithms were evaluated on the MIT-BIH dataset for heart rate anomaly detection. While effective, the study defined variables outside the fixed range (60–100 bpm) as anomalies, which limited its real-world applicability. Local Outlier Factor (LOF) and Random Forest performed best, highlighting the potential of simulated data for training. Park et al. [9] used GANs to generate fraud labels for datasets lacking them, applying logistic regression and XGBoost for classification, with SHAP analysis identifying key features. In [10], unsupervised clustering (K-means and K-medoids) was used to detect anomalies in wearable sensor data,

with K-means slightly outperforming K-medoids. However, dataset details were lacking.

3. SYSTEM ARCHITECTURE

The architecture begins with interconnected medical devices. These devices continuously generate patient-centric data, including vital signs, treatment parameters, and diagnostic readings. Since these devices operate in real-time, they form the foundation of the anomaly detection system by providing continuous streams of healthcare data.

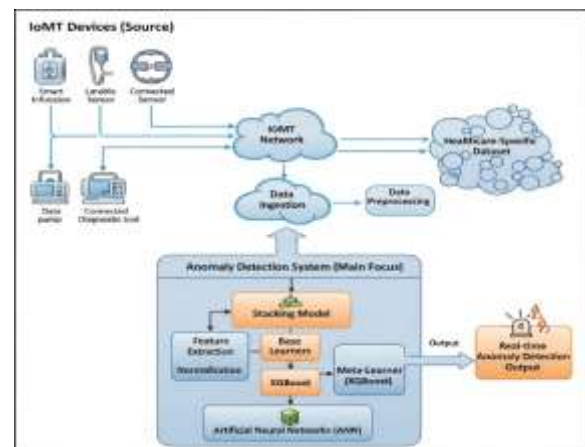


Fig 1: System Architecture

3.1 Methodology

The methodology for this research was designed to create a reliable and efficient anomaly detection framework tailored for Internet of Medical Things (IoMT) environments. Since IoMT networks combine both medical data and network

traffic, the approach needed to capture healthcare-specific patterns while also identifying malicious cyber activities. The process was divided into six main stages: data acquisition, preprocessing, baseline model development, proposed model construction, performance evaluation, and real-time prediction.

3.2 IoMT Network and Data Ingestion:

The data collected from IoMT devices is transmitted through the IoMT network. This network acts as the communication backbone, ensuring that patient information and device activity are securely transferred. The data ingestion module then aggregates these inputs, combining healthcare-specific datasets with network traffic logs. This integration ensures that both medical and cyber contexts are captured, creating a comprehensive dataset for anomaly detection.

3.3 Preprocessing Layer

Once ingested, the data undergoes preprocessing to prepare it for machine learning analysis. This stage includes feature extraction, normalization, and encoding. Feature extraction identifies the most relevant attributes from healthcare and

network datasets. Normalization ensures that all features contribute equally by transforming them into a uniform scale. For example, normalization is expressed as::

$$X = \frac{X - \mu}{\sigma}$$

where X is the original feature, μ is the mean, and σ is the standard deviation. Encoding converts categorical variables such as attack categories into numerical form, making them compatible with supervised learning algorithms.

3.4 Anomaly Detection System (Core Layer) :

The central component of the architecture is the anomaly detection system. It integrates multiple machine learning models to ensure robustness and adaptability. Artificial Neural Networks (ANN) are used to capture complex nonlinear relationships in healthcare and network data. Base learners such as Random Forest, ANN, and XG Boost provide diverse perspectives on anomaly detection. Their outputs are then combined using a meta-learner (XGBoost) in a stacking ensemble model.

The stacking ensemble prediction can be expressed as:

$$\hat{y} = f_{meta}(f_1(x), f_2(x), \dots, f_n(x))$$

where $f_1, f_2, \dots, f_n$ are base learners and f_{meta} is the meta-learner that aggregates their predictions. This layered ensemble ensures high accuracy and adaptability in detecting diverse attack types:

3.5 Real-Time Detection and Output Layer

The final stage of the architecture is the real-time anomaly detection output. Here, the system classifies traffic as either normal or malicious, with healthcare-specific context considered. Alerts are generated instantly, enabling healthcare providers to respond to threats without delay. The outputs can be integrated into hospital dashboards, monitoring systems, or APIs, ensuring seamless deployment in real-world healthcare environments.

4. DESIGN AND CONSTRUCTION

The design of the proposed anomaly detection system follows a structured approach that integrates data engineering, model development, computational architecture, and deployment. The goal is to ensure that the system remains accurate,

efficient, and adaptable to the dynamic nature of IoMT environments.

The first stage of design involves the data acquisition module, which collects heterogeneous inputs from IoMT devices and external datasets. Medical devices such as infusion pumps and wearable sensors generate patient-centric data, while the UNSW-NB15 dataset provides network traffic flows. These sources are combined to form a hybrid dataset that reflects both healthcare and cybersecurity contexts.

The preprocessing pipeline is then constructed to handle real-world data inconsistencies. This includes cleaning duplicate records, imputing missing values, encoding categorical variables, and normalizing numerical attributes. By ensuring data consistency, the preprocessing stage prepares the dataset for effective learning.

The modeling component is designed using a combination of baseline algorithms and the proposed Light GBM classifier. Baseline models such as Random Forest, ANN, and XG Boost establish performance benchmarks, while Light GBM is optimized for efficiency and scalability. Its leaf-wise

tree growth strategy allows deeper learning of discriminative patterns, making it suitable for resource-constrained IoMT devices.

The construction of the system includes a prediction engine that integrates the trained models and generates real-time outputs. This engine classifies traffic as either normal or malicious, with healthcare-specific context considered. A visualization module is also developed to present results through dashboards or APIs, enabling healthcare providers to monitor anomalies instantly

5 RESULTS AND DISCUSSION

The experimental results demonstrate that the proposed hybrid deep learning model effectively identifies multiple IoMT attack categories, including Normal, Worms, Reconnaissance, Fuzzers, Exploits, and DoS.

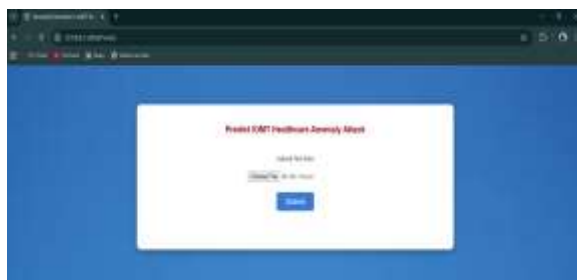


Fig 2: Attack Prediction Page

This page allows the user to upload IoMT healthcare test data for anomaly detection. After submitting the dataset, the trained

stacking model analyses the input and predicts whether the network traffic is normal or an attack.

[illegible]

Fig 3: IoMT Attack Prediction

The model achieves accurate and consistent classification by effectively capturing complex spatial-temporal patterns in network traffic, resulting in reduced misclassification rates. Its performance was evaluated on the UNSW-NB15 and merged UNSW-Healthcare datasets using multiple ML models, with metrics such as accuracy, precision, recall, and F1-score demonstrating the effectiveness of the proposed approach.

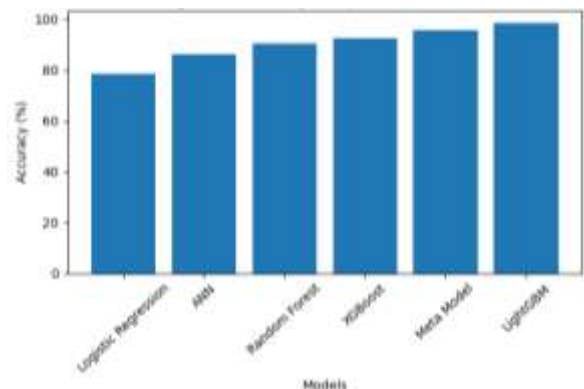


Fig 4: Accuracy Comparison of models

The comparative analysis shows a steady improvement in accuracy from traditional models to advanced ensemble techniques, with the stacking model significantly enhancing detection performance. The proposed Light GBM model achieved the highest accuracy of 98.31%, demonstrating superior efficiency and effectiveness in handling complex IoMT healthcare data.

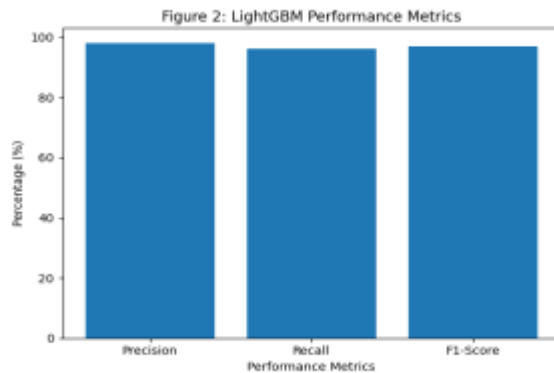


Fig 5: performance Metrics

Figure 5 presents the Precision, Recall, and F1-Score of the Light GBM model. The model achieved:

- **Precision: 98.14%**
- **Recall: 96.13%**
- **F1-Score: 97.08%**

High precision, recall, and F1-score demonstrate that the model reliably detects attacks with minimal false alarms, ensuring

both accuracy and dependability in healthcare environments.

6. CONCLUSION

This research presents an effective anomaly detection framework for IoMT environments by integrating healthcare data with network intrusion information using advanced machine learning techniques. Individual models like ANN showed moderate performance but lacked robustness for complex IoMT data. The proposed Light GBM-based meta model significantly improved detection capability by capturing complex feature interactions. It achieved high performance with 98.31% accuracy, along with strong precision, recall, and F1-score. The integration of healthcare context enhanced detection accuracy and reduced false positives. High recall ensured effective identification of diverse cyber-attacks, minimizing undetected threats. The balanced F1-score demonstrated reliable and consistent model performance. Overall, the proposed approach offers a scalable, accurate, and practical solution for real-time IoMT cyber security in healthcare systems.

FUTURE SCOPE: Future work can focus on integrating real-time data streaming using

platforms like Apache Kafka or MQTT to enable continuous monitoring in live IoMT environments. Advanced models such as Graph Neural Networks and Transformer models can be explored to better capture temporal and relational patterns. Additionally, incorporating automated hyperparameter tuning and explainable AI techniques will improve model accuracy, transparency, and trust for real-world healthcare deployments.

REFERENCES

1. D. V. Dimitrov, "Medical Internet of Things and big data in healthcare," *Healthcare Informat. Res.*, vol. 22, no. 3, p. 156, 2021, doi: 10.4258/hir.2016.22.3.156.
2. P. Manickam, S. A. Mariappan, S. M. Murugesan, S. Hansda, A. Kaushik, R. Shinde, and S. P. Thipperudraswamy, "Artificial intelligence (AI) and Internet of Medical Things (IoMT) assisted biomedical systems for intelligent healthcare," *Biosensors*, vol. 12, no. 8, p. 562, Jul. 2022, doi: 10.3390/bios12080562.
3. W. Sun, Z. Cai, Y. Li, F. Liu, S. Fang, and G. Wang, "Security and privacy in the medical Internet of Things: A review," *Secur. Commun. Netw.*, vol. 2018, pp. 1–9, Jul. 2018, doi: 10.1155/2018/5978636.
4. T. Yaqoob, H. Abbas, and M. Atiquzzaman, "Security vulnerabilities, attacks, countermeasures, and regulations of networked medical devices—A review," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3723–3768, 4th Quart., 2019, doi: 10.1109/COMST.2019.2914094.
5. T. Levy-Loboda, E. Sheetrit, I. F. Liberty, A. Haim, and N. Nissim, "Personalized insulin dose manipulation attack and its detection using interval-based temporal patterns and machine learning algorithms," *J. Biomed. Informat.*, vol. 132, Aug. 2022, Art. no. 104129, doi: 10.1016/j.jbi.2022.104129.
6. P. Sharma, S. Jain, S. Gupta, and V. Chamola, "Role of machine learning and deep learning in securing 5G-driven industrial IoT applications," *Ad Hoc Netw.*, vol. 123, Dec. 2021, Art. no. 102685, doi: 10.1016/j.adhoc.2021.102685.
7. N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf. (MilCIS)*, Nov. 2015, pp. 1–



International Journal of
DATA SCIENCE AND IOT MANAGEMENT SYSTEM

Peer Reviewed, Referred & Indexed Journal

ISSN: 3068-272X

www.ijdim.com

Original Research Paper

-
6. [Online]. Available:
<https://ieeexplore.ieee.org/document/7348942>
 8. E. Šabić, D. Keeley, B. Henderson, and S. Nannemann, "Healthcare and anomaly detection: Using machine learning to predict anomalies in heart rate data," *AI Soc.*, vol. 36, no. 1, pp. 149–158, May 2020, doi: 10.1007/s00146-020-00985-1.
 9. S. Park, K. H. Lee, B. Ko, and N. Kim, "Unsupervised anomaly detection with generative adversarial networks in mammography," *Sci. Rep.*, vol. 13, no. 1, pp. 1–10, Feb. 2023, doi: 10.1038/s41598-023-29521-z.
 10. M. Kavitha, P. V. V. S. Srinivas, P. S. L. Kalyampudi, S. F. Choragudi, and S. Srinivasulu, "Machine learning techniques for anomaly detection in smart healthcare," in *Proc. 3rd Int. Conf. Inventive Res. Comput. Appl. (ICIRCA)*, Sep. 2021, pp. 1350–1356.